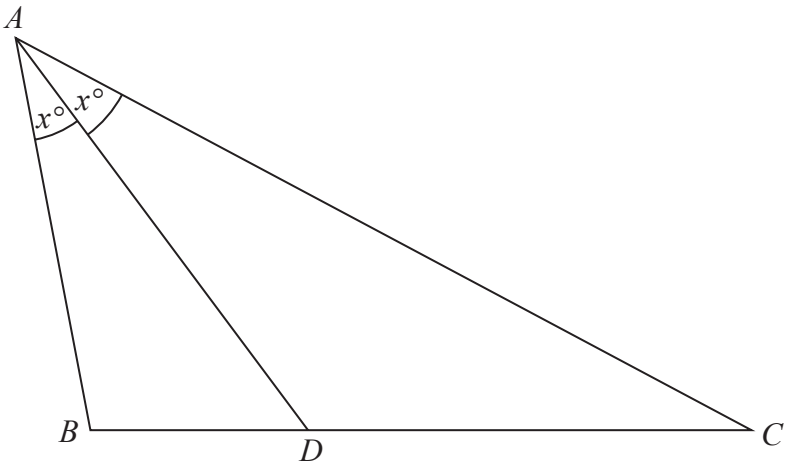
23 ABC is a triangle.



D is the point on BC such that angle BAD = angle $DAC = x°$

Prove that $\frac{AB}{BD} = \frac{AC}{DC}$

Triangles ABD and ADC share altitude h to BC.

Area(ABD)/Area(ADC) = BD/DC.

Area(ABD) = (1/2)AB·AD sin x.

Area(ADC) = (1/2)AC·AD sin x.

Therefore BD/DC = AB/AC.

Cross-rearrange: AB/BD = AC/DC, as required.

1 mark
area of first triangle

1 mark
area of second triangle

1 mark
shared altitude gives BD/DC

1 mark
rearrange to required ratio

(Total for Question 23 is 4 marks)

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